DIRAC: Forensic Model for Measuring Information Entropy in LLM Generated Software

A Mathematical Model to quantify Structural Decay via the Dirac Risk Unit (DRU)

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2026-04-28

1 Introduction

Large Language Models (LLMs) are being rapidly integrated as part of Software Development Life Cycle (SDLC). This can causing friction in system due to probabilistic nature of code generation.

Current metrics like **Lines of Code (LoC)** and **Velocity** fails to capture the invisible debts.

This paper introduces a new forensic framework **D**eterministic **I**ntegrity & **R**isk **A**ssessment **C**ore (**DIRAC**) version 1.0, to measure "Entropy Tax" in AI Engineering.

It applies Shannon Entropy from *Information Theory* to create three novel metrics introduced in this framework which are as follows:

1. **Dirac Risk Unit (DRU)** Υ_D
2. **Scaling Constant** ($\kappa = 0.15$)
3. **Dirac Density Threshold DDT** ($\Theta_D = 10.5$)

Using mathematical model, you will see that every unit of entropy that is added, results in a marginal cost of $8.52 per commits in Technical Debts and Reviews.

Using this framework, I will derive mathematical correlation between **AI Code**, Increased **Tech Debts & Maintenance** and **Financial Leakage**.

2 Archival Notice

DIRAC is now archived. Active research and development have ceased, and no further updates are planned. This paper is retained for historical and reference purposes. For questions about the published work, you can reach me at qubitai.in@gmail.com.